

Twentieth International Linguistics Olympiad

Bansko (Bulgaria), 23–29 July 2023

Individual Contest Problems

Rules for writing out the solutions

Do not copy the problems. Write down your solution to each problem on a separate sheet or sheets. On each sheet indicate the number of the problem, the number of your seat and your surname. If you do not do this, your work may be mislaid or misattributed.

Unless stated differently, you should describe any patterns or rules that you identified in the data. Otherwise your solution will not be awarded full marks.

Problem 1 (20 points). Here are two different forms of some verbs in Guazacapán Xinka and their English translations:

	Form 1	Form 2	Translation
1.	piriy' <i>it saw (it)</i>	imbir'i <i>I see (it)</i>	<i>see</i>
2.	im'ay <i>it said (it)</i>	inim'a <i>I say (it)</i>	<i>say, tell</i>
3.	aplay' etc.	inapal'a etc.	<i>open (it)</i>
4.	k'aniy'	in̥k'an'i	<i>trap</i>
5.	ɬikniy'	in̥ɬikin'i	<i>obey,</i> <i>believe</i>
6.	tundiy'	indunat'i	<i>play (an instrument)</i>
7.	ɬakɬay'	in̥sakats'a	<i>steal</i>
8.	kiɬiy'	in̥gits'i	<i>roast</i>
9.	hik'ay	inhik'a	<i>sew, weave</i>
10.	hiniy'	inhin'i	<i>learn, know</i>
11.	yut̥muy'	inyut̥um'u	<i>suck candy</i>
12.	ip̥lay'	inipal'a	<i>bathe (it)</i>
13.	p'ahn̥iy'	imp'ahan'i	<i>dig</i>
14.	ter'oy	inder'o	<i>kill</i>
15.	netkay'	?	<i>push</i>
16.	kiri̥y'	?	<i>pull</i>
17.	p'uh̥r̥uy'	?	<i>make holes</i>
18.	her'oy	?	<i>smooth out</i>

(a) Fill in the gaps in rows 15–18.

Here are some more verb forms in Guazacapán Xinka and their English translations:

19.	imbiiri?	<i>I was seen</i>
20.	hik'aŋ	<i>I sewed, wove (it)</i>
21.	ɪŋk'ata?	<i>I lie down</i>
22.	kirin'	<i>I pulled (it)</i>
23.	insiiru?	<i>I hurried</i>
24.	inişapa?	<i>I leave</i>
25.	inima?	<i>I am being told // (it) is being said to me</i>
26.	şawşan'	<i>I planted (it)</i>
27.	şur'uŋ	<i>I chose (it)</i>
28.	indeero?	<i>I was killed, I died</i>
29.	imbiri?	<i>I am being seen</i>
30.	heero?	<i>it was smoothed out</i>
31.	p'ihna?	<i>it leaped</i>
32.	şawşa?	<i>it was planted</i>
33.	ak'okama?	<i>it kneels</i>
34.	atero?	<i>it is being killed, it dies</i>
35.	k'aata?	<i>it lay down</i>
36.	asiru?	<i>it hurries</i>
37.	işpa?	<i>it left</i>
38.	aşakaşa?	<i>it is being stolen</i>
39.	her'oŋ	?
40.	inişpa?	?
41.	apla?	?
42.	?	<i>it chose (it)</i>
43.	?	<i>I saw (it)</i>
44.	?	<i>I plant (it)</i>
45.	?	<i>it was trapped</i>
46.	?	<i>I leap</i>
47.	?	<i>it is being sewn, woven</i>
48.	?	<i>I was bathed</i>

(b) Fill in the gaps in rows 39–48.

△ ŋ = *ng* in *hang*. ş ≈ *sh* in *sheesh*. ts ≈ *ts* in *Patsy*. ʔ, h, ɬ, w and y are consonants; ʔ and consonants followed by ' are pronounced by briefly blocking the flow of air in the throat. i is a vowel. A doubled letter denotes length.

The Guazacapán Xinka language belongs to the Xinkan family. It was spoken in the town of Guazacapán in Guatemala. It no longer has any fluent native speakers—in the past 50 years, all former speakers of the language switched to speaking only Spanish instead.

—Samuel Ahmed

Problem 2 (20 points). Here are some sentences in Apurinã and their approximate English equivalents:

- | | |
|-------------------------------------|---|
| 1. nuta sykaru nykanawate uwamukary | } <i>I gave my canoe to her.</i> |
| 2. nykanawate nysykaru uwamukary | |
| 3. pita âтары іpurââ | — <i>You_{sg} drank the water.</i> |
| 4. kyky mynaru nyherêka sytumukary | — <i>The man brought my blood to the woman.</i> |
| 5. herêkatxi uâtary sytu | — <i>The woman drank the blood.</i> |
| 6. nhiximakyte pixinhikary | — <i>You_{sg} remembered my fish (sg).</i> |
| 7. pitxiparyte umynary kykymukary | — <i>She brought your_{sg} banana to the man.</i> |
| 8. hâtakuru xinhikary naikute | — <i>The girl remembered my house.</i> |
| 9. sytu apy ysykanu nutamukary kyky | — <i>The man gave the woman's bone to me.</i> |
| 10. nhîpurââte nâtary | — <i>I drank my water.</i> |
| 11. kywîtxi apukary | — <i>He found the head.</i> |
| 12. ywa mynanu kyky uky nutamukary | — <i>He brought the man's eye to me.</i> |
| 13. ximaky nhikary apîtxi | — <i>The fish (sg) ate the bone.</i> |
| 14. hâtakuru txiparyte napukary | — <i>I found the girl's banana.</i> |
| 15. ximaky kywy inhikary | — <i>He ate the fish's (sg) head.</i> |
| 16. hâtakuru uapukaru | — <i>She found the girl.</i> |

(a) Translate into English:

17. uwa xinhikaru sytu
18. txipary nhinhikary
19. kyky aikute pysykary ywamukary

(b) This sentence is grammatically incorrect. Explain why, and provide its grammatically correct equivalent and its English translation:

20. uwa umynary kanawa ximakymukary

(c) Translate into Apurinã in two ways:

21. *You_{sg} found the house.*
22. *I brought your_{sg} canoe to the girl.*
23. *He brought your_{sg} water to me.*
24. *The man drank the girl's blood.*
25. *I remembered the eye.*
26. *You_{sg} ate my banana.*

⚠ The Apurinã language belongs to the Purus group of the Arawakan family. It is spoken by no more than 30% of a total ethnic population of ca. 9,500 Apurinã along the Purus River, in the northwestern Amazon region of Brazil.

nh ≈ ny in *canyon*; tx = ch in *church*; x = sh in *sheesh*. y = oo in *goose* but with the lips unrounded. â, ê, î are nasal vowels. —Ksenia Gilyarova

Problem 3 (20 points). Here are some sentences in Coastal Marind and their English translations:

1. **ob-dahetok** — *You_{sg} really returned.*
2. **nake-dahetok** — *We returned.*
3. **nambat-ihwin** — *Aw, I cried.*
4. **nambe-otab** — *We really chased you_{pl} away.*
5. **um-kulayawn** — *He laughed at you_{sg} in vain.*
6. **e-keseh** — *You_{pl} spat on him.*
7. **ubate-huyanab** — *Aw, you_{sg} silenced us.*
8. **nakum-hoyab** — *I silenced you_{pl} in vain.*
9. **nambame-yadayawn** — *Oh, we left you_{sg} behind!*
10. **ebe-yadanawn** — *You_{pl} really left us behind.*
11. **ibatum-yadewn** — *Aw, you_{pl} left him behind in vain.*
12. **a-isana** — *He cut me in half.*
13. **obam-eseb** — *Oh, you_{sg} cut him in half!*

The following verbs have a grammatical difference from the verbs above:

14. **bate-tamanab** — *Aw, we floated to the surface.*
15. **ba-tamab** — *You_{pl} really floated to the surface.*
16. **a-yuyayah** — *You_{sg} shuddered.*
17. **bam-yuyanah** — *Oh, I shuddered!*

(a) Translate into English:

18. **e-yuyanah**
19. **ba-ihwin**
20. **nambum-kisayah**
21. **bat-yadawn**
22. **ubamum-kolewn**

(b) Translate into Coastal Marind:

23. *He floated to the surface.*
24. *Aw, you_{sg} returned.*
25. *Oh, I silenced him!*
26. *He really chased us away.*
27. *You_{pl} really spat on us in vain.*

△ The Coastal Marind language belongs to the Anim family. It is spoken by approx. 8,000 people in the South Papua province of Indonesia.

y and **y** are consonants. In the Coastal Marind verb forms, a hyphen (-) is used to separate two halves of a verb that are pronounced as if they were separate words.

—David Hultman (Consultant: Bruno Olsson)

Problem 4 (20 points). Here are some verb forms in Plains Cree and their English translations:

- | | |
|--------------------------------|---|
| 1. nimawinēskomik | — <i>he challenges me</i> |
| 2. kiwāpamināwāw | — <i>you_{pl} see me</i> |
| 3. kinakininān | — <i>you_{sg} stop us, you_{pl} stop us</i> |
| 4. nikakwēcimānānak | — <i>we ask them</i> |
| 5. kiwīchihāw | — <i>you_{sg} help him</i> |
| 6. nikakwēcimānān | — <i>we ask him</i> |
| 7. niwāpamāw | — <i>I see him</i> |
| 8. kimawinēskomin | — <i>you_{sg} challenge me</i> |
| 9. nimawinēskomikonānak | — <i>they challenge us</i> |
| 10. kikakwēcimik | — <i>he asks you_{sg}</i> |
| 11. kiwāpamitinān | — <i>we see you_{sg}, we see you_{pl}</i> |
| 12. ninakināwak | — <i>I stop them</i> |
| 13. kiwīchihikowāw | — <i>he helps you_{pl}</i> |
| 14. kinakinikwak | — <i>they stop you_{sg}</i> |
| 15. kiwāpamāwāwak | — <i>you_{pl} see them</i> |
| 16. kiwīchitin | — <i>I help you_{sg}</i> |

Here are some more verb forms in Plains Cree and their English translations:

- | | |
|----------------------------------|---------------------------------------|
| 17. ē-nakināyēk | — <i>as you_{pl} stop him</i> |
| 18. ē-wīchihak | — <i>as I help him</i> |
| 19. mawinēskomikoyāhkwāwi | — <i>if they challenge us</i> |
| 20. wāpamaci | — <i>if you_{sg} see him</i> |
| 21. wīchitako | — <i>if I help you_{pl}</i> |

(a) Translate into English:

- 22. **ē-wāpamikoyēk**
- 23. **ninakinikonān**
- 24. **kikakwēcimāwāw**
- 25. **kiwīchitināwāw**

(b) Translate into Plains Cree:

- 26. *if we ask them*
- 27. *they challenge you_{pl}*
- 28. *they help me*
- 29. *you_{sg} see them*
- 30. *I stop you_{pl}*

△ The Plains Cree language belongs to the Algonquian branch of the Algic family. It is spoken by approx. 12,000 people in South Saskatchewan and central Alberta in Canada.

In this problem *we* = we without you. **w** and **y** are consonants. The mark ^ː indicates that the vowel is long.

—Lidija Stanovnik, Aleksejs Peguševs

Problem 5 (20 points). Here are some numerals in Supyire and their values:

baashuunni	7
benjaaga na ninjin	21
ŋkwuu taanre na beeshuunni na kajkuro	285
kampwoo na ŋkwuu shuunni na beetaanre na baani	626
kampwɔhii sicyɛɛre na benjaaga na kɛ na baaricyɛɛre	1639

(a) Write in digits:

- kampwɔhii shuunni na kɛ
- ŋkuu na baataanre

(b) Write out in the Supyire language: 15; 109; 152; 403; 1534.

⚠ The Supyire language belongs to the Senufo group of the Atlantic-Congo family. It is spoken by approx. 350,000 people in Mali and Côte d'Ivoire.

The words are given in a simplified transcription. ϵ and $\mathfrak{ɔ}$ are vowels. η and $\mathfrak{p}$ are consonants.

—*Aleksejs Peguševs*

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Good luck!